

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and Orchestrate”

Project Tile – AI Interview Trainer Agent

Domain of Project – Education / Career Development

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Problem Statement

Problem Statement : Interview Trainer Agent

The Challenge

Preparing for job interviews is often difficult because candidates lack personalized guidance, structured preparation plans, realistic practice opportunities, and timely feedback. Interview questions, HR expectations, behavioral scenarios, communication strategies, and company-specific interview patterns are scattered across multiple sources. Without personalized preparation and assessment, many candidates struggle to identify weaknesses and improve their interview performance.

The Objective

Build an AI-powered Interview Trainer Agent that helps candidates prepare for technical and non-technical interviews through personalized coaching, mock interviews, answer evaluation, and readiness assessment.

Build an AI Interview Trainer Agent that:

- Analyzes resumes

- Conducts mock interviews

- Evaluates answers

- Generates readiness reports

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used IBM watsonx Orchestrate + IBM Granite Models + RAG

Proposed Solution

Proposed Solution: AI Interview Trainer Agent

The proposed solution is an AI-powered Interview Trainer Agent built using IBM watsonx Orchestrate, IBM Granite Models, and Retrieval-Augmented Generation (RAG).

Key capabilities of the Interview Trainer Agent:

- Resume-Based Personalized Preparation
- Resume Analysis
- Technical Interview Preparation
- HR Interview Preparation
- Soft Skills Coaching
- Mock Interview Assessments
- Answer Evaluation
- Final Readiness Reports

The platform uses specialized agents coordinated by the **Interview Trainer Agent** to provide personalized interview coaching.

The solution leverages a multi-agent architecture where specialized agents collaborate to analyze resumes, retrieve interview knowledge, generate role-specific questions, evaluate responses, and provide actionable feedback.

By combining Agentic AI and RAG, the platform delivers personalized interview preparation that helps candidates improve confidence, communication, and interview readiness.

Orchestrate Components Used

Core Agent

- **Interview Trainer Agent** - Central orchestration agent.

Specialized Agents

- **Resume Analyzer Agent** - Resume parsing and candidate profiling.
- **Technical Interview RAG Agent** - Retrieves interview knowledge from the knowledge base.
- **Question Generator Agent** - Generates role-specific and fallback interview questions.
- **HR Coach Agent** - HR and Behavioral interview preparation.
- **Soft Skills Coach Agent** - Communication, leadership, teamwork, and self-introduction coaching.

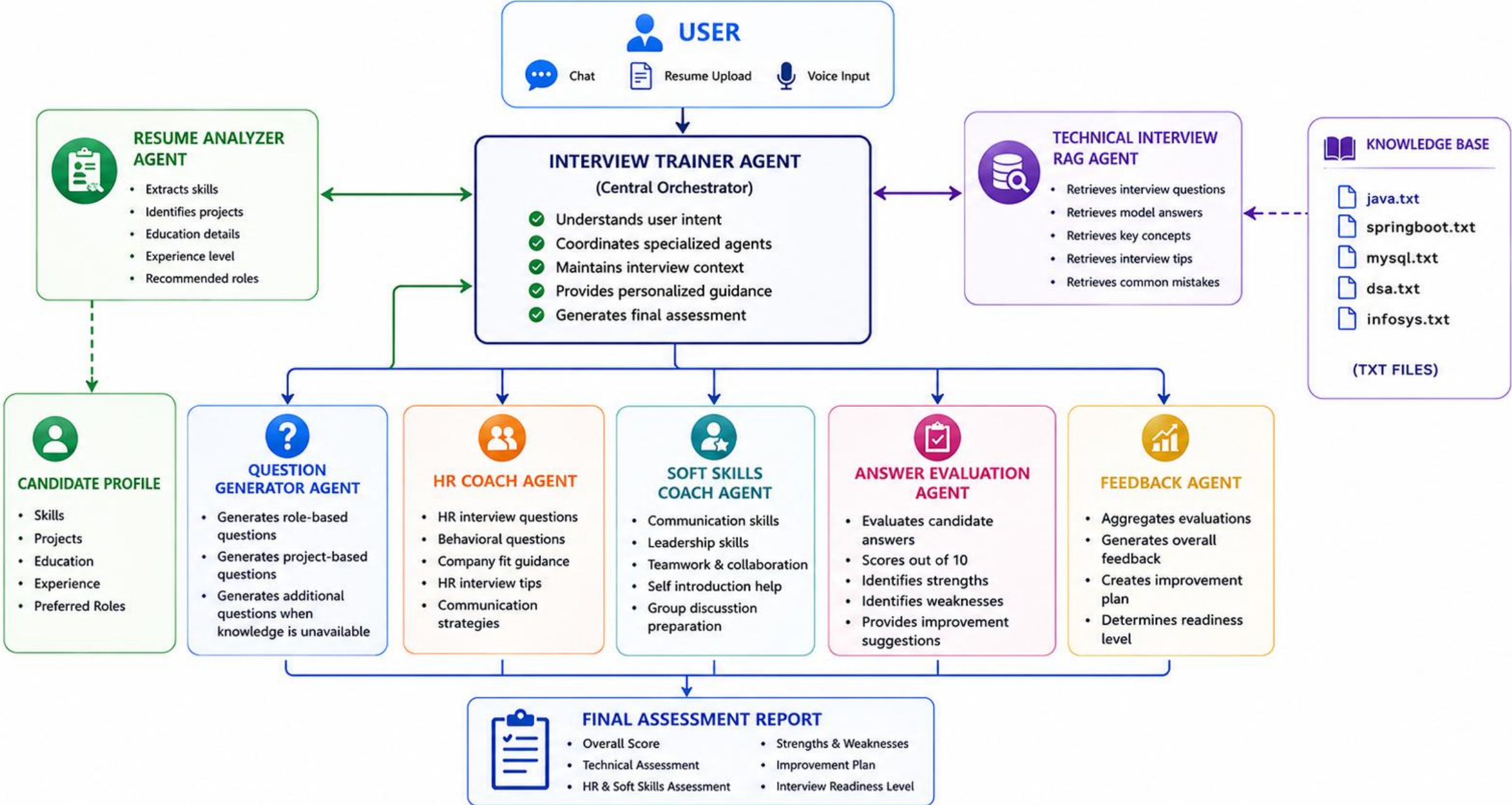
Evaluation Agents

- **Answer Evaluation Agent** - Scores answers and identifies improvement areas.
- **Feedback Agent** - Generates final interview assessment reports.

IBM Services

- **Voice Configuration** - IBM Watson Speech-to-Text & IBM Watson Text-to-Speech
- **Knowledge Base** - Stores interview preparation datasets and reference content.

Architecture Blueprint



Interview Trainer Workflow



Outcome

A comprehensive interview readiness report with scores, strengths, weaknesses and personalized improvement recommendations to help candidates succeed in their interviews.

Orchestrate Code – after deployment

Agent successfully embedded using IBM watsonx Orchestrate Web Chat SDK.

IBM watsonx Orchestrate

EnglishContact SalesBuy Now

16 trial days left

?

GP

Agent chat / Manage agents / Interview Trainer Agent

Test agentliveDeploy

Interview Trainer Agent

Model : GPT-OSS 120B – Ope...

Build with AITalk to agent

Running in debug mode

Profile

Knowledge

Toolset

Behavior

Channels Preview

Embedded agent

DraftLive

Make sure you deploy your agent after updating the Voice configuration

English | English - United States, Ethan (Male) - Natural

Embed on your website

Test your agent easily. Copy and paste this snippet into your environment to preview before deploying. [Learn more](#)

```
<script>
window.wx0Configuration = {
  orchestrationID: "ed557a7693944eb6a94a6558f1a5e639_bf1c6820-5a61-43b1-838a-6",
  hostURL: "https://au-syd.watson-orchestrate.cloud.ibm.com",
  rootElementID: "root",
  deploymentPlatform: "ibmcloud",
  crn: "crn:vi:bluemix:public:watsonx-orchestrate:au-syd:a/ed557a7693944eb6a94",
  chatOptions: {
    agentId: "2ba0b90f-3b8a-43c3-9ce9-b56f57c193f0",
    agentEnvironmentId: "fb7b924c-1b18-438d-9980-acb07db098bf",
  }
};
setTimeout(function () {
  const script = document.createElement('script');
  script.src = `${window.wx0Configuration.hostURL}/wxochat/wxoLoader.js?embed=
script.addEventListener('load', function () {
  wxoLoader.init();
});
document.head.appendChild(script);
}, 0);
</script>
```

9:28 PM

Hello, AI Interview Trainer

Accuracy of generated answers may vary. Please double-check responses.

Technical Skills Development

Company Specific Questions

Role-based Interview Questions

Type something...

Role of Agentic AI in the solution

How Agentic AI Powers the Interview Trainer Agent

Unlike a traditional chatbot, the **Interview Trainer Agent** operates as an intelligent multi-agent system that can analyze, reason, plan, and act autonomously to support interview preparation.

Agentic AI Responsibilities

- **Resume Understanding** - Extracts skills, projects, education, and experience from uploaded resumes.
- **Knowledge Retrieval** - Retrieves interview questions, concepts, model answers, and company-specific content from the RAG knowledge base.
- **Adaptive Question Generation** - Generates personalized interview questions based on role, skills, and experience level.
- **Answer Evaluation** - Scores candidate responses and identifies strengths, weaknesses, and missing concepts.
- **Personalized Coaching** - Provides technical, HR, behavioral, and soft-skill guidance tailored to the candidate profile.
- **Final Readiness Assessment** - Produces interview readiness reports, improvement plans, and preparation recommendations.

Benefits of Agentic AI

- ✓ Resume-based personalization
- ✓ Context-aware interview preparation
- ✓ Dynamic question generation
- ✓ Automated answer evaluation
- ✓ Continuous feedback loop
- ✓ End-to-end interview readiness assessment
- ✓ Multi-agent collaboration and orchestration

Key Value

Agentic AI transforms interview preparation from a static question-answer process into an intelligent, personalized coaching experience that adapts to each candidate's profile and performance.

Technology Used

- **IBM Cloud Platform** - Provides the secure execution environment for IBM watsonx Orchestrate services and agent deployment.
- **IBM watsonx Orchestrate** - Multi-agent orchestration platform used to build, manage, and coordinate specialized interview agents.
- **GPT-OSS 120B Foundation Model** - Generates interview questions, model answers, answer evaluations, coaching recommendations, and final assessments.
- **Retrieval-Augmented Generation (RAG)** - Retrieves interview questions, concepts, model answers, company-specific preparation material, and interview guidance from the knowledge base.
- **IBM Watson Speech-to-Text** - Converts candidate voice responses into text for interview processing and evaluation.
- **IBM Watson Text-to-Speech** - Converts interview questions, coaching feedback, and assessments into spoken responses.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance.
- **Knowledge Base** - Stores interview preparation content in TXT format: java.txt, springboot.txt, mysql.txt, dsa.txt, and infosys.txt.
- **Multi-Agent Architecture** - Enables collaboration between:
 - Interview Trainer Agent
 - Resume Analyzer Agent
 - Technical Interview RAG Agent
 - Question Generator Agent
 - HR Coach Agent
 - Soft Skills Coach Agent
 - Answer Evaluation Agent
 - Feedback Agent

Novelty and Uniqueness

The **AI Interview Trainer Agent** leverages **Agentic AI, Retrieval-Augmented Generation (RAG), Resume Intelligence, and Voice Interaction** to deliver a personalized and end-to-end interview preparation experience.

Key Innovations

- **Multi-Agent Architecture** where specialized agents collaborate for resume analysis, interview preparation, answer evaluation, and feedback generation.
- **Resume-Driven Personalization** that tailors interview questions, preparation plans, and coaching based on candidate skills, projects, and experience level.
- **RAG-Powered Knowledge Retrieval** using curated interview content, technical concepts, model answers, and company-specific preparation materials.
- **Voice-Enabled Interview Experience** through IBM Watson Speech-to-Text and Text-to-Speech services, enabling realistic interview practice.
- **Automated Answer Evaluation** with scoring, strengths, weaknesses, missing concepts, and actionable improvement suggestions.
- **Comprehensive Interview Readiness Assessment** covering technical, HR, behavioral, and soft-skill preparation in a single platform.

Unique Value Proposition

Unlike traditional interview preparation platforms that provide static question banks and generic guidance, our solution functions as an **intelligent AI interview coach** that adapts to each candidate's profile, performance, and career goals. This helps candidates build confidence, improve communication skills, and increase their chances of success in competitive hiring processes.

Git Hub Link

GitHub Repository

<https://github.com/Gowtham-gangster/AI-Interview-Agent>

Repository Includes:

- README.md
- Architecture Documentation
- Knowledge Base Files
- Project Presentation
- Embedded Agent HTML
- Screenshots

Demo Results – Part 1

Resume Analysis

You 12:11 PM

Analyze Resume

PDF Gowtham Resume

Interview Trainer Agent 12:11 PM | Show Reasoning

Resume Analysis Report

Name: Pusuloori Gowtham

Technical Skills:

- Programming Languages: Java, Python, C
- Web Development: HTML, CSS, JavaScript, Responsive Design, UI Development
- Back-End: Node.js, Basic Java backend development
- Databases: MySQL, SQL, PostgreSQL
- Tools & Platforms: Git, GitHub, VS Code, IntelliJ IDEA, Jupyter Notebook

Projects:

- Mask-Aware Face Recognition (Dec 2025 – Mar 2026) – Python, OpenCV, basic ML preprocessing for masked/unmasked face detection.
- Smart Prescription & Medicine Reminder (Oct 2025 – Dec 2025) – Full-stack web app (HTML/CSS/JS, Node.js, MySQL) with backend APIs, user dashboard, and medication reminder logic.

Education:

- B.Tech – Computer Science & Engineering – MLR Institute of Technology, Hyderabad (Sept 2023) – CGPA 8.97/10
- Intermediate (Class XII) – Sri Chaitanya Junior College (Jun 2021 – May 2023) – CGPA 96.9%
- SSC (Class X) – Sri Chaitanya High School (Jun 2020 – May 2021) – CGPA 10.0/10

Experience Level: Fresher (recent graduate with solid project & internship-type experience)

Recommended Roles:

- Web Development Intern
- Junior Front-End Developer
- Entry-Level Full-Stack Developer
- Entry-Level JavaScript/Node.js Developer

Strengths

- Strong foundation in core programming (Java, Python, C)
- Hands-on full-stack project experience using modern web technologies
- Good command of version control (Git/GitHub) and IDEs
- Excellent academic record

Potential Improvement Areas

- Deepen expertise in front-end frameworks (React, Angular, Vue)
- Advance back-end skills (Spring Boot, RESTful services, micro-services)

Type something...

Technical Preparation

You 12:14 PM

Prepare me for Java Developer Fresher interview

Interview Trainer Agent 12:14 PM

Java Developer (Fresher) Interview Preparation

Tailored to your background (B.Tech CSE, strong Java basics, project experience, OCI certification)

1

Core Topics to Master

Category	Key Sub-topics	Why It Matters
Core Java	- OOP concepts (inheritance, polymorphism, abstraction, encapsulation) - Data types, Collections (List, Set, Map) - Exception handling (checked vs unchecked, custom exceptions) - Generics, Streams API, Lambda expressions - Multithreading (Thread life-cycle, <code>synchronized</code>, <code>volatile</code>, <code>ExecutorService</code>) - Java 8+ features (Optional, method references)	Most interview questions are built around these fundamentals.
JDBC & SQL	- Connection, Statement, PreparedStatement - Transaction management (commit, rollback) - ResultSet handling - Common SQL queries (JOIN, GROUP BY, sub-queries)	Your project used MySQL; be ready to discuss data access code.
Web Basics	- HTTP methods, status codes, REST principles - JSON handling (Jackson/Gson) - Basic servlet lifecycle (if asked)	Shows you understand how Java fits into web apps.
Build & Version Control	- Maven/Gradle basics (pom.xml, dependencies) - Git commands (branch, merge, rebase)	Expected for any Java role.
Testing	- JUnit basics, assertions, test annotations - Mockito for mocking (conceptual)	Demonstrates quality-first mindset.
Design Patterns (light)	Singleton, Factory, DAO, MVC	Frequently asked to gauge architectural thinking.
OCI Cloud (optional)	Core OCI services you know (Compute, Object Storage)	Your OCI certification can be a differentiator.

2

Frequently Asked Questions & Model Answers

#	Question	Model Answer (concise, 1-2 min)
1	Tell us about yourself.	"I am Pusuloori Gowtham, a recent B.Tech graduate in Computer Science from MLR Institute of Technology with a CGPA of 8.97. I have solid Java fundamentals and have built two full-stack projects: a mask-aware face-recognition system using Python/OpenCV and a smart prescription web app using HTML/CSS/JavaScript, Node.js, and MySQL. I am also OCI-certified, which gave me exposure to cloud fundamentals. I'm eager to start my career as a Java developer, applying my programming, problem-solving, and quick-learning abilities."
2	Explain OOP concepts with a Java example.	"OOP stands for Object-Oriented Programming. It comprises four pillars: Encapsulation (bundling data & methods; e.g., a <code>BankAccount</code> class with private <code>balance</code> and public <code>deposit/withdraw</code>), Inheritance (a subclass inherits fields/methods; e.g., <code>SavingsAccount</code> extends <code>BankAccount</code>), Polymorphism (method overriding – <code>withdraw</code> behaves differently in

Type something...

Demo Results – Part 2

Mock Interview

You 12:22 PM

Start a mock interview.



Interview Trainer Agent 12:28 PM

Final Mock-Interview Assessment – HR / Behavioral (Fresher Java Developer)

Question	Score (/10)	Key Strengths	Areas for Improvement
1. Explain technical concept to non-technical audience	8	Strong analogy, visual aid, audience-centric, clear outcome	Add quantitative result, preparation details, brief reflection
2. Group project role & contribution	7	Clear role, concrete technical work, team coordination, result	Include metrics, describe collaboration process, reflect on learning
3. Difficult Java bug (ConcurrentModificationException)	8	Precise bug identification, systematic debugging, correct solution, successful fix	Mention test metrics, alternative approaches considered, lesson learned
4. Rapidly learning OCI cloud	8	Structured 3-step learning plan, certification, direct impact on deployment	Provide timeframe, knowledge-transfer to team, quantify benefit, add reflection
5. Prioritising tasks & time-management	8	Uses Eisenhower Matrix, Pomodoro, GitHub issue tracking, focuses on critical path, delivers ahead of schedule	Could cite a specific example of a deadline met early, mention any tools (e.g., calendar) for further completeness

Overall Score: 7.8/10 (average of the five questions)

Strengths Summary

- Clear Communication** – Consistently uses analogies, visual aids, and structured explanations that make complex ideas understandable.
- Problem-Solving Rigor** – Demonstrates systematic debugging, root-cause analysis, and methodical solution design.
- Self-Directed Learning** – Shows ability to acquire new technologies quickly through focused study, hands-on sandboxing, and certification.
- Organised Workflow** – Applies proven time-management frameworks (Eisenhower Matrix, Pomodoro) and uses GitHub for tracking, ensuring deadlines are met.
- Team Collaboration** – Takes initiative in repository management, assists teammates, and acts as a communication bridge.

Weaknesses / Gaps

- Quantitative Impact:** Most answers lack specific numbers (e.g., accuracy improvement, time saved, deadline reduction) that make achievements more tangible.
- Reflection:** Few responses end with a concise takeaway or learning point; adding this shows self-awareness and growth.
- Timeline Details:** For rapid-learning scenarios, stating the exact time taken (days/weeks) strengthens the impression of efficiency.
- Team Knowledge Transfer:** When acquiring new skills, mention how you shared that knowledge with peers (e.g., mini-workshop, documentation).

Personalized Improvement Plan

Action	How to Implement	Target Completion
Add metrics to stories	Review each STAR story; quantify results (e.g., "accuracy rose from 68% → 85%", "deployment time cut by 30%", "completed certification in 10 days").	Next mock interview
Include a reflection line	After each answer, add a 1-sentence lesson learned or how you'll apply it in future roles.	Ongoing
Practice concise timelines	When describing rapid learning, state the exact period ("within 2 weeks").	Before next interview
Show knowledge transfer	Mention any documentation, demo, or quick training session you gave to teammates after learning OCI.	Next relevant answer

Interview Trainer Agent 12:22 PM | Show Reasoning ▾

Mock Interview – HR / Behavioral Questions (Fresher Java Developer)

We will go through 5 questions, one at a time.

After each answer I will evaluate it (score/10, strengths, weaknesses, improvement suggestions) and then move to the next question.

Question 1

Can you describe a time during your studies or a project where you had to explain a technical concept to someone without a technical background? How did you ensure they understood?

(Please answer in about 1–2 minutes. When you're ready, type your response.)

You 12:24 PM

During my work on the Smart Prescription & Medicine Reminder web application, I had to present our system architecture to a panel of non-technical evaluators who needed to understand how our software operated. To make the concept accessible, I explained how our Node.js backend APIs and MySQL database functioned together by using a simple restaurant analogy. I compared the backend API to a waiter who takes an order from a customer at a table, delivers it to the kitchen cook to retrieve the requested information, and then brings it back to the customer. I completely avoided database jargon like relational queries, foreign keys, or schemas, and instead described the database as a securely locked, digital filing cabinet that organizes medicine data into individual user folders. To ensure they fully understood, I paired this analogy with a basic, color-coded visual diagram tracking the path of a single medicine reminder from the user's screen to the server. This communication strategy successfully bridged the technical gap, as the evaluators were able to comprehend our full-stack workflow and even asked follow-up questions regarding how the system handles multiple data requests simultaneously.

Interview Trainer Agent 12:24 PM

Answer Evaluation – Question 1

Score: 8/10

Strengths

- Clear Analogy:** Used a relatable restaurant analogy that effectively mapped backend API and database concepts to everyday experiences.
- Audience-Focused:** Recognized the need to avoid technical jargon and tailored language to non-technical evaluators.
- Visual Aid:** Mentioned a color-coded diagram, showing use of multiple communication channels.
- Outcome:** Demonstrated that the audience understood and engaged with follow-up questions.

Weaknesses / Missing Points

- Specific Impact:** Could quantify the result (e.g., "the panel gave us a top rating" or "the presentation helped secure project approval").
- Personal Role Detail:** Briefly mentioned "I presented," but didn't elaborate on preparation steps (e.g., rehearsing, gathering feedback).
- Reflection:** No mention of what you learned or how you would improve the explanation in future.

Improvement Suggestions

- Add Metrics:** Include a concrete outcome (e.g., "received a 9/10 score from the evaluators").
- Detail Preparation:** Briefly describe how you prepared the analogy and diagram (e.g., "drafted the diagram on a whiteboard, practiced the story with a peer").
- Reflect:** End with a short insight (e.g., "I realized that visual metaphors greatly enhance comprehension, so I now start every stakeholder demo with a simple analogy").

Question 2

Tell us about a group project you worked on in college. What role did you play, and how did you contribute to the team's success?



Type something...



Type something...



Future Scope

1. Company-Specific Interview Preparation

Integrate company-specific interview datasets (TCS, Infosys, Accenture, Wipro, Amazon, Microsoft, Google, etc.) to provide targeted preparation based on the candidate's desired organization.

2. Real-Time Voice Mock Interviews

Expand voice-enabled interviews using IBM Watson Speech-to-Text and Text-to-Speech to simulate realistic interview environments and improve communication skills.

3. Coding & Technical Assessment Integration

Support live coding challenges, system design interviews, and automated code evaluation for software engineering roles.

4. Personalized Learning Roadmaps

Generate adaptive preparation plans based on candidate performance, weaknesses, and target job roles.

5. Interview Analytics Dashboard

Provide detailed progress tracking, performance trends, readiness scores, and improvement recommendations through an interactive dashboard.

6. Multi-Language Support

Enable interview preparation in multiple languages to improve accessibility for candidates from diverse backgrounds.

7. Enterprise & Campus Deployment

Extend the platform for colleges, training institutes, and recruitment teams to conduct large-scale interview preparation and assessments.

Thank You!

Thank you for your time and interest.